



Testing Infrastructure as Code Using OpenTofu and Localstack

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Who is Pieter?



CLOUD NATIVE
COMPUTING FOUNDATION

The cycle

```
(.venv) → 5-tests git:(main) ✗ tofu validate  
Success! The configuration is valid.
```

```
(.venv) → 5-tests git:(main) ✗ tofu show -json tfplan | tf-summarize
```

+-----+-----+	
CHANGE RESOURCE	
+-----+-----+	
add	aws_s3_bucket.this
+-----+-----+	
	aws_s3_bucket_policy.this
+-----+-----+	

Plan: 2 to add, 0 to change, 0 to destroy.

Do you want to perform these actions?

OpenTofu will perform the actions described above.

Only 'yes' will be accepted to approve.

Enter a value: yes

aws_s3_bucket.this: Creating...

Error: creating S3 Bucket (pieters-bucket): operation error S3: CreateBucket, https response error StatusCode: 400, RequestID: 644d7757-04bf-440e-b5d3-5c3c265389d8, HostID: s9lzHYrFp76ZVxRcpX9+5cjAnEH2R0uNkd2BHfIa6UkFVdtjf5mKR3/eTPFvsiP/XV/VLi31234=, api error MalformedXML: The XML you provided was not well-formed or did not validate against our published schema

```
with aws_s3_bucket.this,  
on main.tf line 1, in resource "aws_s3_bucket" "this":  
  1: resource "aws_s3_bucket" "this" {
```

Plan: 2 to add, 0 to change, 0 to destroy.

Do you want to perform these actions?

OpenTofu will perform the actions described above.
Only 'yes' will be accepted to approve.

Enter a value: yes

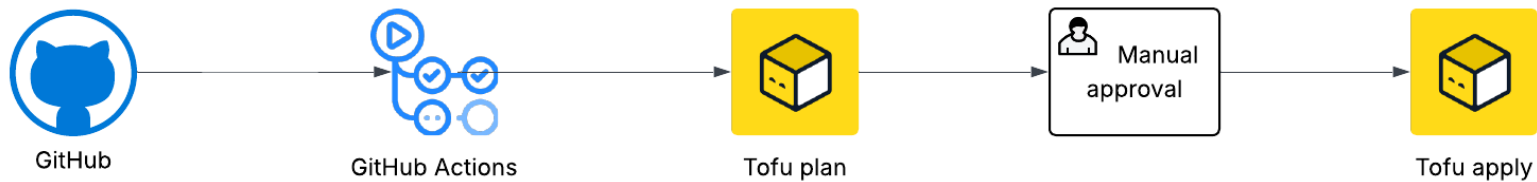
aws_s3_bucket.this: Creating...

Error: creating S3 Bucket (pieters-bucket): operation error S3: createBucket, api error MalformedXML: The XML you provided was not well-formed or did not validate against our published schema, response error StatusCode: 400, RequestID: 644d7757-04bf-440e-b5d3-5c3c265389d8, HostID: s9lzHYrFp76ZVxRcpX9+5cjAnEH2R0uNkd2BHFia6UKfvutjT5mKR3/eTPFvs134=, api error MalformedXML: The XML you provided was not well-formed or did not validate against our published schema

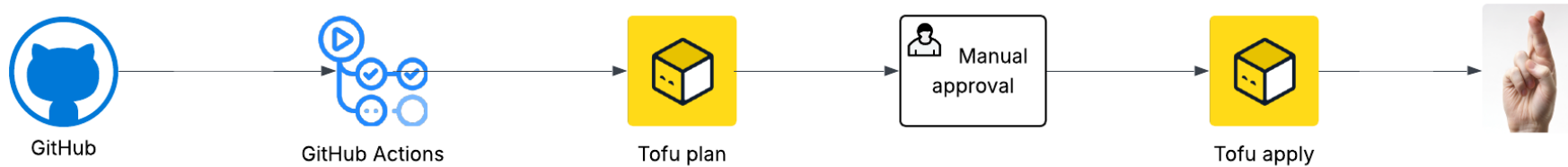
with aws_s3_bucket.this,
on main.tf line 1, in resource "aws_s3_bucket" "this":
1: resource "aws_s3_bucket" "this" {



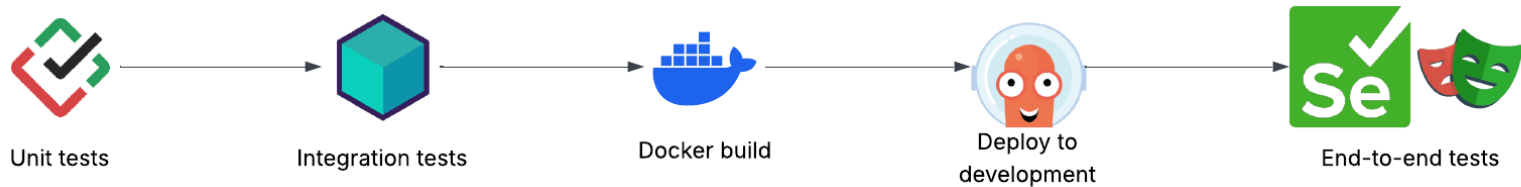
Infrastructure provisioning



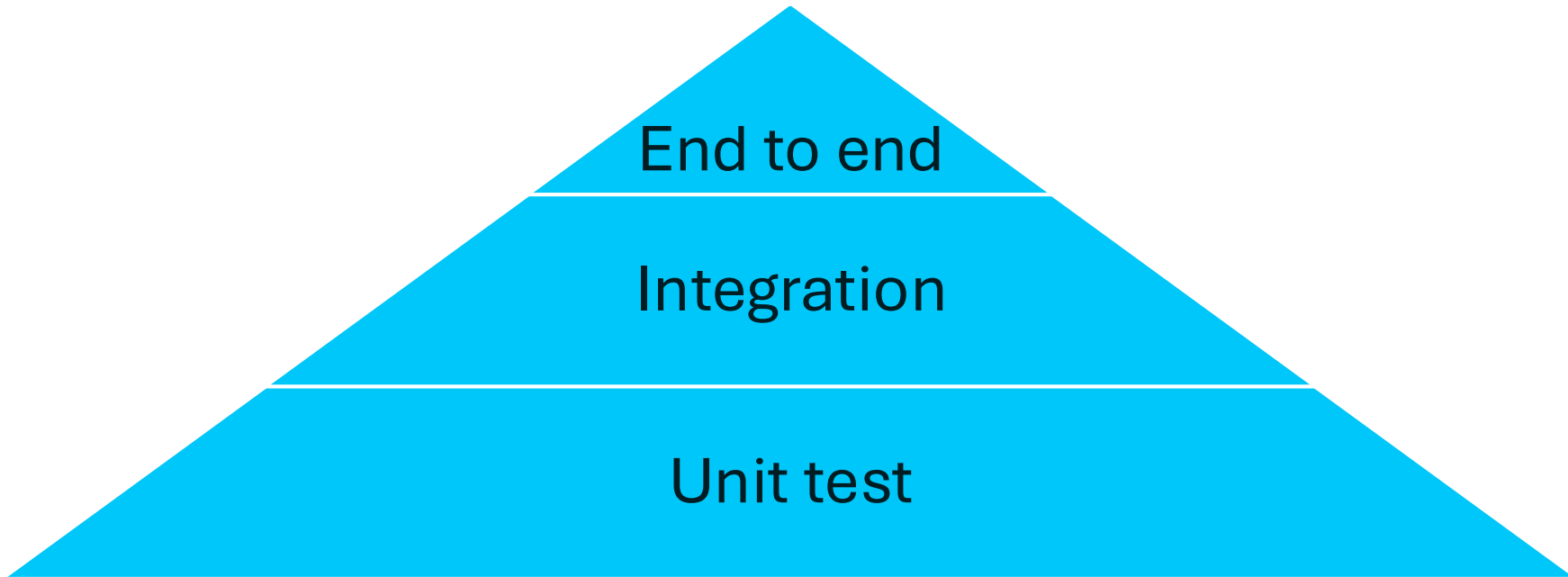
Infrastructure provisioning



Java container builds



Testing pyramid



Terminology

- **Orchestration repository:** Repository with IaC code that is deployable as-is towards an environment.
- **Module repository:** A module is a reusable component. This module should provide a building block solution (combine multiple services and resources). They should be opinionated.

Unit test

Validate inner workings of module

Given

- Dependencies
- Input variables

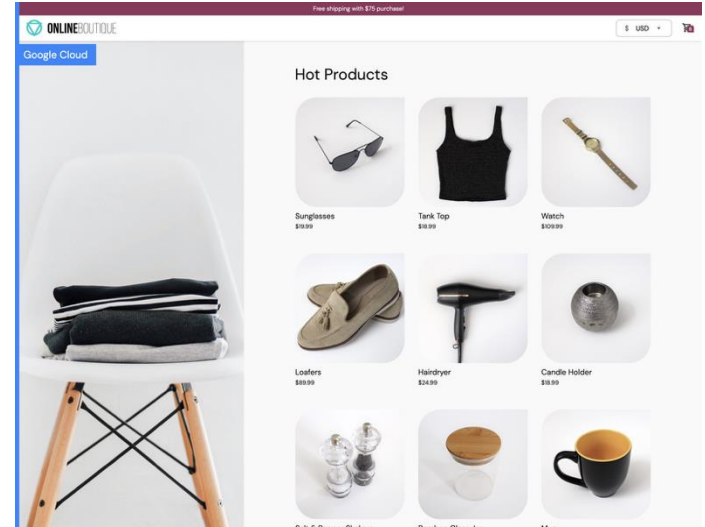
When

- Plan or apply

Then

- Output variables
- Deployed configuration

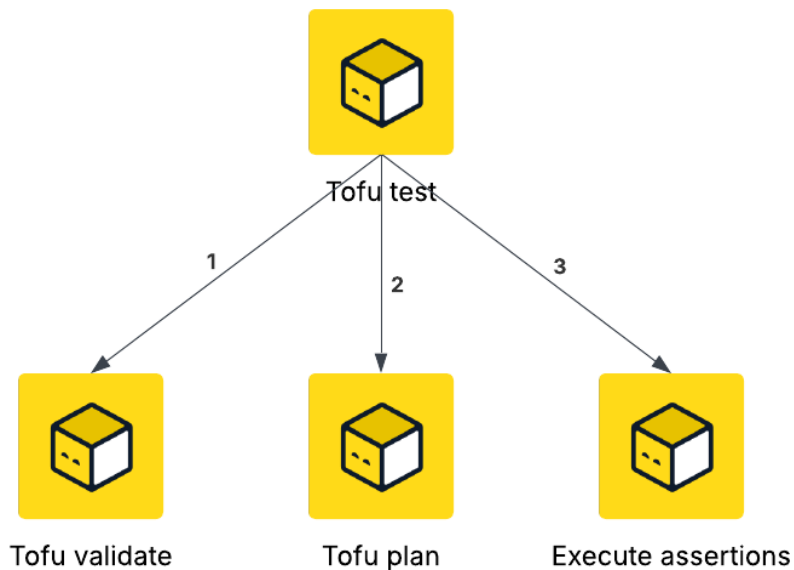
Deployment validation & End-to-end tests



Demo time

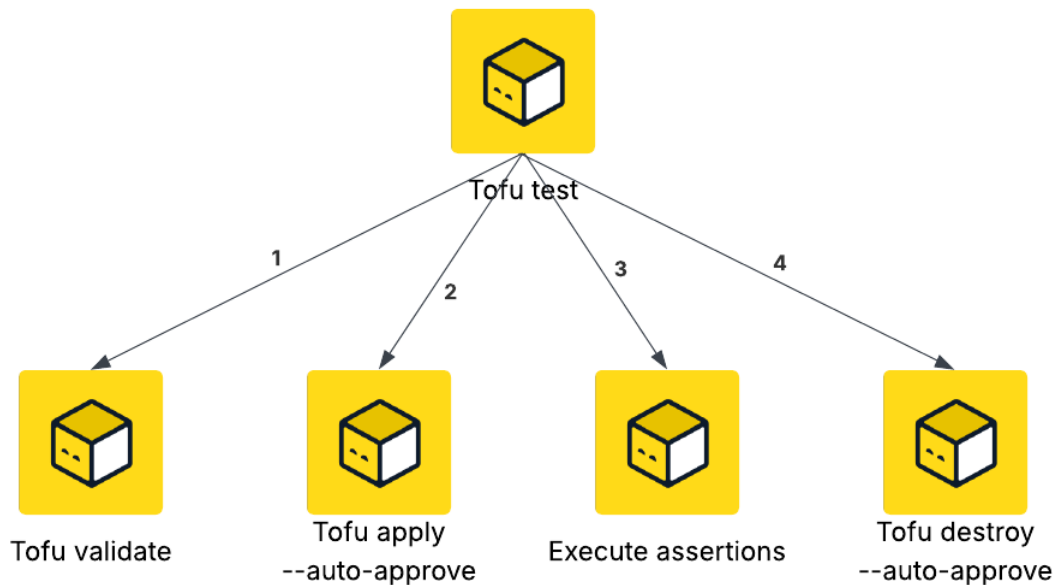
Terraform testing: mode plan

Behind the scenes

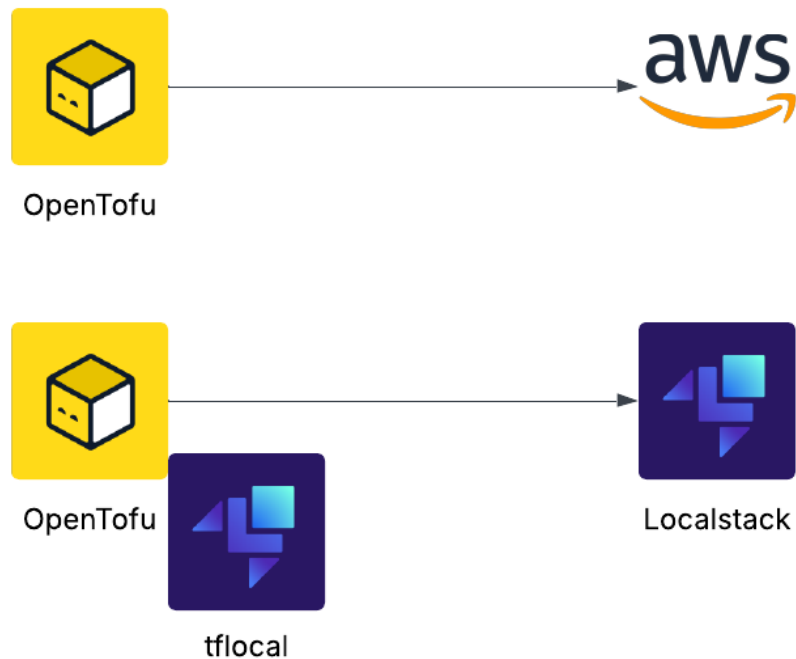


Terraform testing: mode apply

Behind the scenes



Localstack



GetMoto

Localstack alternative

- Boto SDK mock
- Has REST API implementation
- Docker Support
- Less “like AWS”
 - Side-effects are not implemented (good or bad?)
- (With some workarounds) Works with tflocal and OpenTofu tests
- <https://github.com/getmoto/moto>



GH repo



Blog



Q&A



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